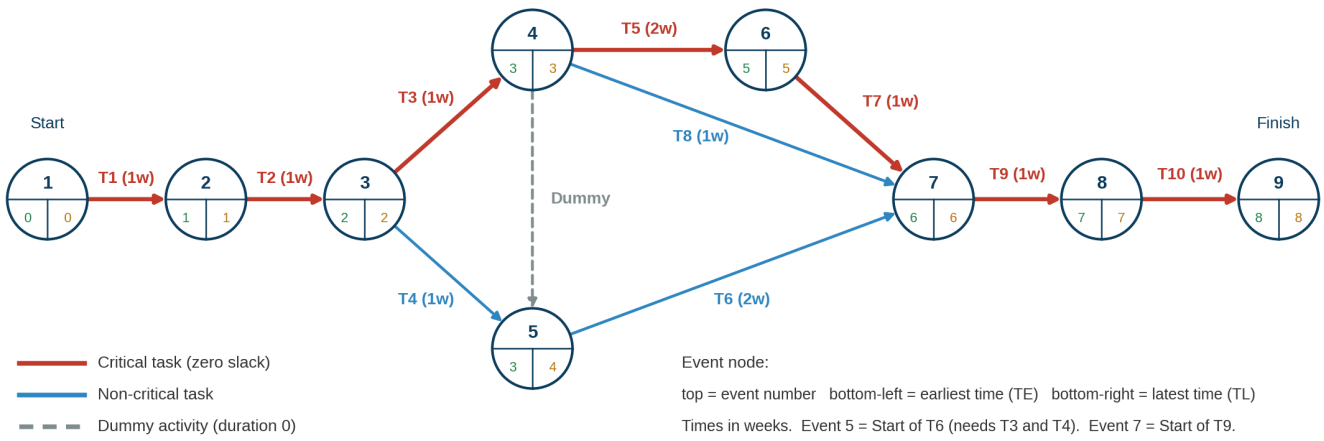
# WATER SUPPLY COMPLAINT & BILLING SYSTEM - ACTIVITIES TABLE

Tasks schedule derived from System Request and Feasibility Study Analysis:

| Task Code | Task Description | Duration (Weeks) | Predecessors |
|---|---|---|---|
| **T1** | Feasibility Study & System Request Approval | 1 | - |
| **T2** | Requirements Elicitation & Complaint / Billing Workflow Analysis | 1 | T1 |
| **T3** | System Architecture & Database Design | 1 | T2 |
| **T4** | UI/UX Design for Customer Portal / App & Management Dashboard | 1 | T2 |
| **T5** | Backend Development: Complaint Engine (Tracking No., Auto-Routing, Duplicate Merging) & Billing Module | 2 | T3 |
| **T6** | Customer Portal & Mobile App Development | 2 | T3, T4 |
| **T7** | SMS / E-bill & Online Payment Gateway Integration | 1 | T5 |
| **T8** | Security, Data Protection & Existing Customer / Meter Data Integration | 1 | T3 |
| **T9** | System Integration Testing & User Acceptance Testing (UAT) | 1 | T6, T7, T8 |
| **T10** | Staff Training & Pilot Deployment (one branch first) | 1 | T9 |

## PERT NETWORK DIAGRAM




## Critical Path Analysis:

**Critical Path** = T1 → T2 → T3 → T5 → T7 → T9 → T10
**Minimum Total Project Duration** = 1 + 1 + 1 + 2 + 1 + 1 + 1 = **8 Weeks** (56 days)

**All paths compared:**

**Path 1:** T1 → T2 → T3 → T5 → T7 → T9 → T10 = 1+1+1+2+1+1+1 = **8 weeks (longest)**

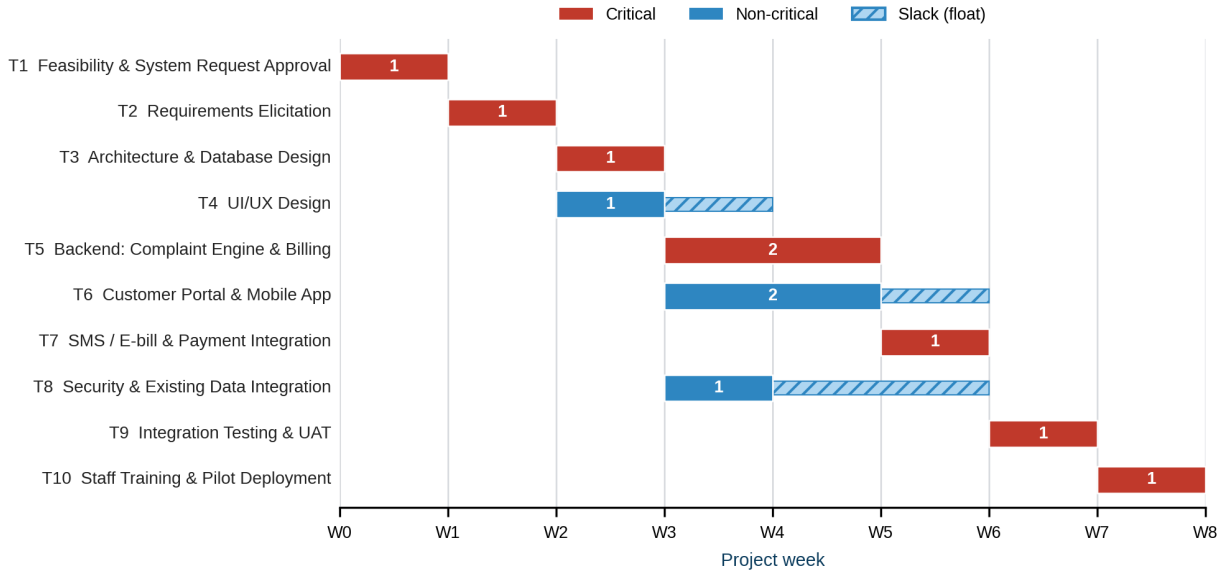
**Path 2:** T1 → T2 → T3 → T6 → T9 → T10 = 1+1+1+2+1+1 = 7 weeks

**Path 3:** T1 → T2 → T4 → T6 → T9 → T10 = 1+1+1+2+1+1 = 7 weeks

**Path 4:** T1 → T2 → T3 → T8 → T9 → T10 = 1+1+1+1+1+1 = 6 weeks

# WATER SUPPLY COMPLAINT & BILLING SYSTEM - GANTT CHART

Timeline mapping across 8 total project execution weeks:




| Task | Earliest Start | Earliest Finish | Latest Start | Latest Finish | Slack (weeks) | Critical? |
|---|---|---|---|---|---|---|
| **T1** | 0 | 1 | 0 | 1 | 0 | Yes |
| **T2** | 1 | 2 | 1 | 2 | 0 | Yes |
| **T3** | 2 | 3 | 2 | 3 | 0 | Yes |
| **T4** | 2 | 3 | 3 | 4 | 1 | No |
| **T5** | 3 | 5 | 3 | 5 | 0 | Yes |
| **T6** | 3 | 5 | 4 | 6 | 1 | No |
| **T7** | 5 | 6 | 5 | 6 | 0 | Yes |
| **T8** | 3 | 4 | 5 | 6 | 2 | No |
| **T9** | 6 | 7 | 6 | 7 | 0 | Yes |
| **T10** | 7 | 8 | 7 | 8 | 0 | Yes |

Times are measured in weeks from the project start (W0). Slack = Latest Start − Earliest Start.

## Summary & Feasibility Alignment:

**Operational Feasibility:** Each complaint gets a tracking number and is routed automatically to the right technical team. Short staff training (T10) and a pilot at one branch first reduce resistance to leaving paper procedures.

**Technical Feasibility:** A modular web application (HTML, CSS, JavaScript, Java or Python back-end, relational database) plus a mobile app; SMS and online-payment gateways are added through standard APIs (T7). T8 covers integration with existing customer and meter data and the protection of customer data.

**Economic Feasibility:** One-time cost of about 975,700 EGP (including 10% contingency) against benefits of about 3,295,000 EGP per year at full adoption (60% realized in Year 1). Payback is within the first year (about 0.7 years).

**Schedule Constraint:** The critical path of 8 weeks (56 days) fits the 1 January deadline. If the project starts on 1 November 2026, it finishes on about 27 December 2026, leaving roughly 5 days of buffer.